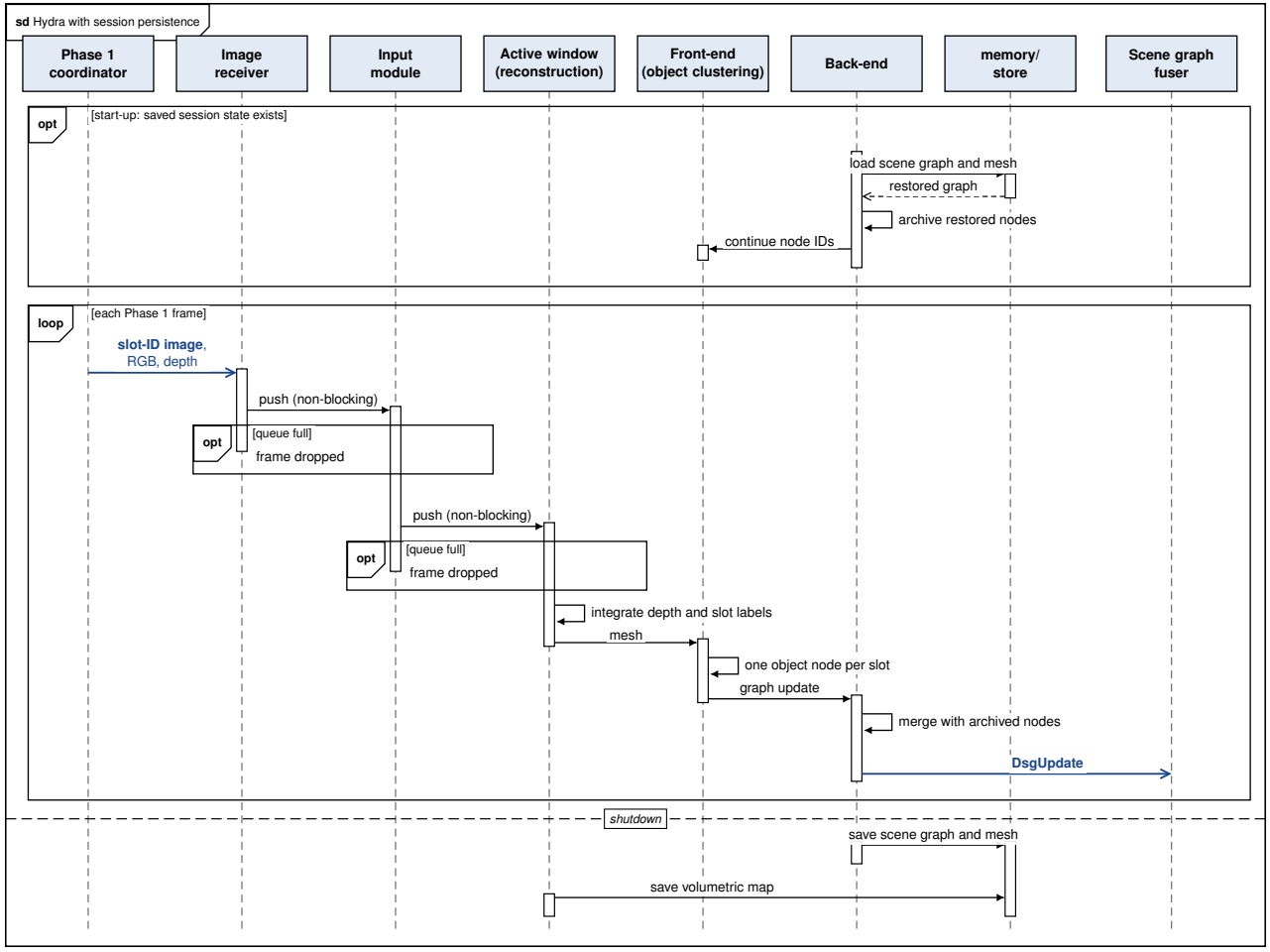
sd Hydra with session persistence
Phase 1 coordinator
Image receiver
Input module
Active window (reconstruction)
Front-end (object clustering)
Back-end
memory/ store
Scene graph fuser
opt
[start-up: saved session state exists]
load scene graph and mesh
restored graph
archive restored nodes
continue node IDs
loop
[each Phase 1 frame]
slot-ID image, RGB, depth
push (non-blocking)
opt
[queue full]
frame dropped
push (non-blocking)
opt
[queue full]
frame dropped
integrate depth and slot labels
mesh
one object node per slot
graph update
merge with archived nodes
DsgUpdate
shutdown
save scene graph and mesh
save volumetric map